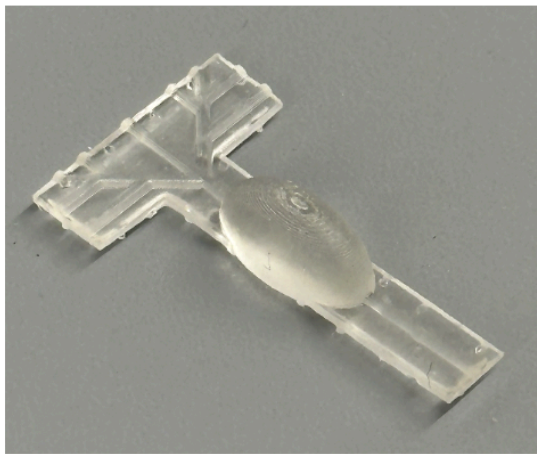


Prototyping Evolution:

The stepwise progression from **V1** to **V3**, along with alternative concepts, was driven by both functional requirements and manufacturing constraints.

- **V1 - external microfluidic layer:** In the earliest version, the microchannels were printed on the outer surface of a silicone teat. This allowed rapid prototyping, but the delicate network tore easily during support removal and use. The external layer also created a raised surface that could be uncomfortable for the infant and harder to clean. These drawbacks motivated embedding the channels within the teat.



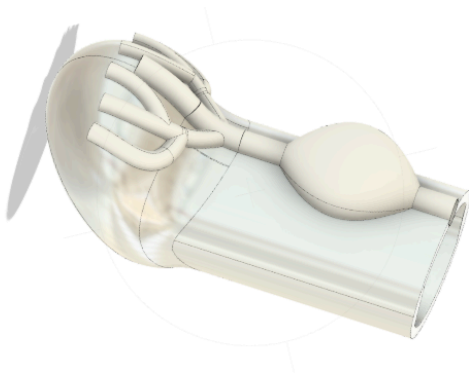
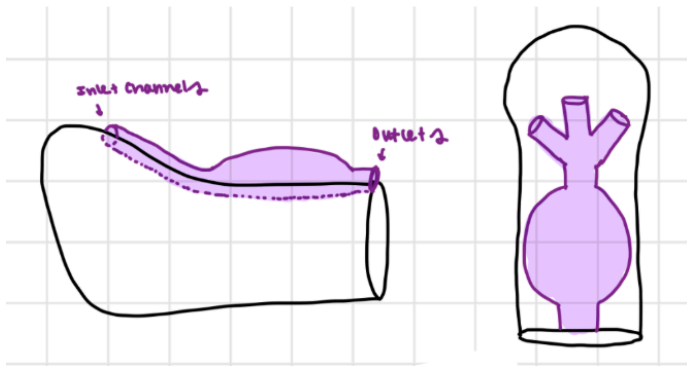
- **V2 - 50:50 embedded design using flexible 80A resin:** To improve durability, the team embedded half of the microfluidic network within the teat. This design reduced tearing and improved fit, but the extremely narrow channels (3 mm OD, 1.8mm ID) frequently collapsed during printing. The position of the reservoir at the very tip of the teat also limited sample collection. These issues highlighted the need for thicker walls and a relocated reservoir. The goal was to maximise saliva capture while maintaining anatomical compatibility.



Design Decision - Avoiding Lingual Placement:

During this stage, the team also evaluated possible mounting positions for the microfluidic device, including the lingual and palatal sides of the teat. Lingual placement was intentionally avoided early on because saliva secreted from the sublingual and submandibular glands is predominantly mucous, with a higher viscosity and a distinct enzyme and buffer profile that could interfere with pH measurement and disrupt laminar flow. Collecting saliva from the palatal side, in contrast, yields more serous fluid reflective of gastric reflux composition and reduces contamination from mucous secretions. This anatomical choice ensured smoother sample transport, greater diagnostic reliability, and consistent flow dynamics for future iterations. [Physiology, Salivation - StatPearls - NCBI Bookshelf](#)

- **V3 - thicker walls and centralised reservoir:** Learning from V2, the third prototype thickened the walls to prevent collapse and moved the reservoir farther down the teat. Supports were removed after post-curing to reduce tearing. This was the first iteration that printed successfully without collapsing channels. However, ongoing prototype testing and simultaneous simulations are being conducted to validate inlet height and reservoir volume. Preliminary observations from early prints suggest that the inlets may still sit too low relative to the palate and that the reservoir might need to be enlarged to ensure consistent sensor contact.



The evolution of the prototypes demonstrates how fabrication limitations (minimum printable channel size), anatomical fit, and mechanical durability informed successive design choices. Each iteration addressed specific shortcomings from the previous version while preserving the core goal of passive, comfortable saliva collection. This iterative prototyping approach ensures that the final design is both functional and manufacturable.

New Design Innovations:

Based on prior simulations, prototypes and feedback, several major changes were introduced:

1. **Teat-integrated microfluidic device:** The millifluidic network is now fully integrated within the pacifier teat rather than being laminated externally. This preserves a smooth exterior surface, reduces choking hazards, and improves fit under the palate. The final geometry places channels along the roof of the teat with a widened reservoir near the distal end.
2. **Reduced number of inlets:** Instead of multiple small inlets, the updated design uses fewer but larger inlets. Larger openings simplify manufacturing, reduce the risk of clogging by viscous saliva, and still provide sufficient sampling area. Fewer inlets also free space for additional structural support.
3. **Increased inlet diameter:** The diameter of each inlet and connecting channel was increased to improve capillary-driven flow and to meet manufacturing tolerances of resin 3-D printing. Wider inlets collect more saliva per unit time and reduce shear stresses that could cause fluid stagnation.
4. **Wider and deeper reservoir:** Enlarging the reservoir ensures that enough sample contacts the pH sensor and provides damping to mitigate pulsatile flow. A deeper chamber also helps maintain contact during variations in sucking intensity.
5. **Addition of separating walls between inlets:** To maximise saliva collection and minimise mixing of bubbles or air pockets, small walls are being added between the inlets at the entrance region. These walls help channel saliva toward the reservoir and reduce dead space; they also increase structural stiffness of the thin roof of the teat.
6. **Biomechanical and FDA Considerations:** To ensure GERDSense meets pacifier safety standards and anatomical requirements, the design integrates biomechanical insights with regulatory guidelines. We use a one-piece silicone teat with a firm guard and ventilation holes, as required by 16 CFR Part 1511 and AAPD guidelines aapd.orggovinfo.gov. Finite-element analyses show that even 2 mm vertical displacements can alter palatal pressure patterns, so the millifluidic layer is thin and compliant to deform naturally under suction. Ultrasound studies of breastfeeding place the nipple tip about 8.1 ± 0.3 mm from the hard-soft palate

junction, guiding us to situate the reservoir near this deepest point without impeding natural sucking [pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov). Hard components such as the pH sensor are housed externally to avoid choking hazards and comply with protrusion limits, leaving only flexible channels inside the infant's mouth.

